Supplement: case selection, replay and audit trail

Barry W. Brook*[1]

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1. University of Tasmania, Hobart, Tasmania, Australia

Corresponding author: Barry W. Brook

S1. Case selection

The initial candidate catalogue is `papers/infrastructure-audit/evidence/claims.tsv`: 189 candidate findings, including positive controls and intended behaviors, compiled read-only on 25 October 2026 from the program's database, package and reference catalogues. Later program findings were added to the pool, including cases 11 and 12. The twelve cases were chosen on 8 November 2026 (`case-selection.md`) by five rules:

1. consequence for inference before defect status;
2. one case may hold several ledger rows;
3. coverage of the finding classes, including positive controls;
4. disclosure must be feasible;
5. findings of little consequence, and anything whose description could cause harm, are left out.

Four reserves were named in case a case failed replay: RevBayes's disabled fossilized birth–death range process, the optimal linear estimation interval's undercoverage in one package, a join error in a scraped diversity table, and a silent identity rotation in a reconstruction library. None was needed.

S2. Replay and independent check, by case

Each case was replayed on 8 November 2026 from pinned inputs. The replay records are in `papers/infrastructure-audit/evidence/replay-20261108/`, in four parts:

- `main/`: cases 1 and 11, and the parent checks;
- `data/`: cases 2, 3, 4 and 10;
- `software/`: cases 5, 6, 8 and 9;
- `calibration/`: cases 7 and 12.

Each record states:

- the required inputs and their hashes;
- the replay command;
- the checks made by a different actor, which are specific to each case:
  - a fresh raw-data calculation;
  - a fresh pinned implementation;
  - reaggregation of retained outputs;
  - a mathematical or control check.

The `basis` column of the claims ledger names the route for each proposition, and the `limit` column states what it does not show.

What counts as independent. Independence is check-specific. The replays were made by Codex's replay team (the main Codex session and bounded GPT-6.1 Sol delegates), combined with separate aggregation, interpolation, geometry or scoring code where the receipt records it. Not every proposition was rederived by a fully separate route: the INDITEK forward run reused the original pinned harness, and only its scoring was separately implemented. Some reviewers used the same model family and shared inputs. The checks are not independent biological datasets, and they are not evidence that using two model families helps.

Availability. Some required inputs are retained locally rather than published: two content-pinned container images, retained posterior logs and source archives. A cold rebuild from public inputs alone was not validated.

Software environment. R 4.6.1 with admtools 0.7.0, fossilbrush 1.0.6 and Rextinct 1.1.0; pyGPlates 1.0.0; content-pinned container images for the software runner. Full versions are recorded in each replay record.

Case notes. Details that the main text cites to this section:

- **Case 3 (Triton).** The row key and the source-priority recipe are defined in `evidence/replay-20261108/data/result` Python and the original R deletion memberships agree exactly.
- **Case 7 (calibration controls).** In the retained 20,000-row tables, the G1 birth–death mean and survival and the G2 Poisson-sampling moments agree with their analytic expectations; the paleobuddy and FossilSim dispersions are 0.9942762 and 0.9903592. The declared overall agreement gate (E4) failed, and the executed hiatus windows were shifted one Myr from the declaration; in that arm 728 cases were excluded, leaving 1,272.
- **Case 9 (plate models).** The TorsvikCocks2017 model's authors document a hybrid mantle frame (anchor plate 0) and a paleomagnetic frame (anchor plate 1); the replay's rotation routine uses anchor 0. The observed route difference is consistent with a frame difference; the archived service's parameters were not established.

- **Case 11 (printed probability).** Four routes: the original direct exact-rational upper-tail sum and R (Codex reference session, 6 November), a complement recurrence and R (main Codex, 8 November), and integer-polynomial convolution (a bounded Codex reviewer).
- **Case 12 (INDITEK).** The forward replay reused the original pinned harness at the three proposal states; the model-free scorer was written separately.

S3. Statements withdrawn or narrowed during replay

Case	Original statement	Found by	Replaced by	Record
3	1,454 changed means, 423 by at least one degree; two per-bin class flips; 11–13% sample inflation	Codex replay, 8 Nov 2026	1,388 changed means, 411 by at least one degree; class flips as in C045	`evidence/replay-20261108/dat`
4	Sample count including a header row	Codex parent count	61,862 samples	`evidence/replay-20261108/mai`
5	"Wrong times above every hiatus"	Codex replay	Errors in affected interpolation segments only	C096 limit
7	PyRate intervals "too narrow"	Codex replay	Undercoverage with upward-shifted intervals; width comparisons do not support "too narrow" and do not establish calibrated widths	C107 limit
10	5,291 damaged histories	Codex replay	2,967 altered histories; 5,291 is the count of cells in scientific notation	C035 limit
1	Records described as "withheld"	Codex replay	Absent from a dated public snapshot; cause not established	C191 limit
5	First draft joined the sparse-model count with the dense-model maximum	Codex wording review, 8 Nov 2026	Each control reported separately	`docs/audits/2026-11-08-wordi` W01
12	First draft attributed the INDITEK archive to Cermeño et al. (2022) and said predictions "replay exactly"	Codex wording review	Herrero-Gascón et al. (2026) synthetic proof example; vectors within about 5×10^{-14}	W02
7	First draft said the PyRate intervals "are not too narrow"	Codex wording review	Width calibration not established either way	W03
all	First draft said every proposition was rederived by a separate route, and that anyone can rerun every case	Codex wording review	Check-specific independence; locally retained inputs named	W07, W08

S4. Reproduction

From the repository root:

```
python3 papers/infrastructure-audit/tools/claims.py check \
  papers/infrastructure-audit/manuscript/claims.csv \
  papers/infrastructure-audit/manuscript/text/{main,supplement,brief,for-agents}.md
```

Each ledger row carries its own `reproduce` command and the commit at which its evidence was verified. Third-party archives are not redistributed; each replay record names the source and its hash, so a reader can obtain and verify the same input. Records absent from public database snapshots are reported as counts only and are not reproduced here.

References